5.19 Solution: Assume that the axis of the cylinder is on the z-axis and the top and the bottom of the cylinder are located at $z = \pm L/2$. Since $\mathbf{M} = M\hat{z}$ is a constant inside the cylinder, $\nabla \cdot \mathbf{M} = 0$. Then, we can find the magnetic scalar potential, using Eq. (5.100), as

$$\begin{aligned}
\Phi_m &= \frac{1}{4\pi} \oint \frac{\mathbf{M}\cdot\mathbf{n}}{|\mathbf{x}-\mathbf{x}'|} da' \\
&= \frac{M}{4\pi} \int_0^{2\pi} d\phi' \int_0^a \rho' d\rho' \left[\frac{1}{\sqrt{\rho^2 + \rho'^2 - 2\rho\rho'\cos(\phi-\phi') + (z-L/2)^2}} \right. \\
&\qquad \left. - \frac{1}{\sqrt{\rho^2 + \rho'^2 - 2\rho\rho'\cos(\phi-\phi') + (z+L/2)^2}} \right].
\end{aligned}$$

On the z-axis, $\rho = 0$, the scalar potential becomes

$$\begin{aligned}
\Phi_m &= \frac{M}{2} \int_0^a \left[\frac{1}{\sqrt{\rho'^2 + (z-L/2)^2}} - \frac{1}{\sqrt{\rho'^2 + (z+L/2)^2}} \right] \rho' d\rho' \\
&= \frac{M}{2} \left[\sqrt{\rho'^2 + \left(z - \frac{L}{2}\right)^2} - \sqrt{\rho'^2 + \left(z + \frac{L}{2}\right)^2} \right] \Bigg|_{\rho'=0}^{a} \\
&= \frac{M}{2} \left[\sqrt{a^2 + \left(z - \frac{L}{2}\right)^2} - \sqrt{a^2 + \left(z + \frac{L}{2}\right)^2} - \left|z - \frac{L}{2}\right| + \left|z + \frac{L}{2}\right| \right] \\
&= \begin{cases}
\frac{M}{2} \left[\sqrt{a^2 + \left(z - \frac{L}{2}\right)^2} - \sqrt{a^2 + \left(z + \frac{L}{2}\right)^2} + L \right], & z > \frac{L}{2} \\[2ex]
\frac{M}{2} \left[\sqrt{a^2 + \left(z - \frac{L}{2}\right)^2} - \sqrt{a^2 + \left(z + \frac{L}{2}\right)^2} - 2z \right], & |z| < \frac{L}{2} \\[2ex]
\frac{M}{2} \left[\sqrt{a^2 + \left(z - \frac{L}{2}\right)^2} - \sqrt{a^2 + \left(z + \frac{L}{2}\right)^2} - L \right], & z < -\frac{L}{2}
\end{cases}.
\end{aligned}$$

The magnetic field is given by

$$\mathbf{H}_{\text{in}} = -\nabla\Phi_m = \frac{M}{2} \left[\frac{z + L/2}{\sqrt{a^2 + (z+L/2)^2}} - \frac{z - L/2}{\sqrt{a^2 + (z-L/2)^2}} - 2 \right] \hat{z},$$

inside the cylinder, and

$$\mathbf{H}_{\text{out}} = -\nabla\Phi_m = \frac{M}{2} \left[\frac{z + L/2}{\sqrt{a^2 + (z+L/2)^2}} - \frac{z - L/2}{\sqrt{a^2 + (z-L/2)^2}} \right] \hat{z},$$

outside the cylinder. The magnetic induction inside and outside the cylinder are the same,

$$\mathbf{B}_{\text{in}} = \mathbf{B}_{\text{out}} = \frac{\mu_0 M}{2}\left[\frac{z + L/2}{\sqrt{a^2 + (z + L/2)^2}} - \frac{z - L/2}{\sqrt{a^2 + (z - L/2)^2}}\right]\hat{z}.$$

At the top and the bottom surfaces of the cylinder, the magnetic induction are the same, with the value

$$\mathbf{B}_{\text{top}} = \mathbf{B}_{\text{bottom}} = \frac{\mu_0 M}{2}\frac{L}{\sqrt{a^2 + L^2}}\hat{z}.$$